

The Over-Dispersed Skellam (ODS) Reserving Model

A Signed Stochastic Framework for Incurred Chain-Ladder Uncertainty

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Abstract

Case-reserve releases and recoveries sometimes produce negative incremental incurred development, motivating predictive aggregate non-life reserving models that allow future incurred movements in both directions. We develop an over-dispersed Skellam (ODS) predictive layer around a selected incurred chain-ladder (CL) completion. Each cell is a signed negative-binomial magnitude and the aggregate has the Skellam mean–variance form in the quasi-likelihood sense, while it follows a distribution of the difference of independent negative-binomial totals. The fitted layer preserves the selected CL mean in every future cell. We propose Mack-MSEP and absolute-value Pearson dispersion calibrations. Mack-MSEP calibration matches the fitted aggregate process variance to the Mack MSEP target, matching one aggregate moment rather than the full predictive distribution. In a 30-by-30 synthetic experiment, empirical 95% interval coverage was 88.9% for the process-only benchmark, 95.7% for Mack-MSEP, and 100.0% for the absolute-value Pearson method.

Keywords: claims reserving, incurred triangles, chain-ladder, mean squared error of prediction, over-dispersed Skellam model, prediction uncertainty, negative incremental claims.

1 Introduction

Stochastic reserving methods supplement a deterministic reserve estimate with a predictive distribution ([England and Verrall, 2002, 2006](#); [Wüthrich and Merz, 2008](#); [Scarth *et al.*, 2020](#)). For incurred triangles, case-reserve reassessments, recoveries, and settlements below previously established case estimates can produce negative incremental incurred development. We study the aggregate future incurred development R , conditional on the observed triangle and centered on a selected incurred chain-ladder (CL) completion. Our motivation is purely practical. In several monthly and

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quarterly incurred triangles, negative implied IBNR balancing amounts occur alongside releases of established case reserves, for example when claims settle below previously reserved amounts. This motivates a predictive distribution that can represent future incurred movements in either direction. Mapping this quantity to payment projections or reported but not settled (RBNS) and incurred but not reported (IBNR) amounts requires additional claims and accounting information.

The over-dispersed Skellam (ODS) model is a predictive layer around a selected CL completion. Its individual cells are signed negative-binomial magnitudes with mean μ_{ij} and variance $\varphi|\mu_{ij}|$, where $\varphi > 1$ is the dispersion parameter. For fixed cell means and dispersion, let M_+ and M_- be the totals of positive and absolute negative future cell means. Then

$$\mathbb{E}[R] = M_+ - M_-, \quad \text{Var}(R) = \varphi(M_+ + M_-).$$

Thus, in the quasi-likelihood mean–variance sense, the aggregate has the Skellam mean–variance structure with variance inflated by φ . Under the full finite-dispersion specification, R is exactly a difference of independent negative-binomial totals. The one-sided sign restriction applies to the fitted parameters and the population specification remains unrestricted. The Mack-MSEP calibration matches the fitted aggregate process variance to the Mack MSEP target, a match of one aggregate second moment, while the Pearson-type calibration uses observed-cell residuals.

Signed incurred development has been modeled using loglinear and linear CL models, shifted-lognormal models, and quasi-likelihood methods (Verrall and Li, 1993; de Alba, 2006; Renshaw and Verrall, 1998). This paper adds an explicit signed discrete predictive distribution around a selected incurred CL mean surface, together with aggregate dispersion calibrations. ODS assumes conditionally independent signed negative-binomial cells, whereas Mack specifies accident-year independence and conditional moments for cumulative development paths (Mack, 1994). A recent stochastic lift of the generalized Cape Cod (GCC) method derives an analytical MSEP for a different mean estimator (Richman and Wüthrich, 2026). The present focus is the signed discrete predictive distribution for incurred increments.

The paper makes three contributions. First, it defines a fitted ODS distribution around a selected incurred CL completion and preserves its mean in each future cell. Second, it specifies Mack-MSEP and absolute-value Pearson dispersion calibrations, while distinguishing aggregate variance matching from matching a full predictive distribution. Third, it compares predictive intervals in a controlled simulation.

2 The model

We define the deterministic incurred CL anchor first and then place the ODS predictive layer around it. The notation also records the observed upper triangle and the future lower triangle.

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2.1 Triangle and incurred CL anchor

Let X_{ij} be incremental incurred development for accident period $i = 1, \dots, I$ and development period $j = 0, \dots, J$. At valuation time I , define

$$\begin{aligned}\mathcal{I}_I &= \{(i, j) : 1 \leq i \leq I, 0 \leq j \leq J, i + j \leq I\}, \\ \mathcal{L}_I &= \{(i, j) : 1 \leq i \leq I, 0 \leq j \leq J, i + j > I\}.\end{aligned}$$

The set $\mathcal{I}_I$ is the observed upper triangle. The set $\mathcal{L}_I$ is the future lower triangle. For each accident period define the last observed development age

$$j_i^* := \min\{J, I - i\}, \quad i = 1, \dots, I.$$

Note that in the square-design special case, $j_i^* = I - i$. The observed data are $\mathcal{D}_I = (X_{ij})_{(i,j) \in \mathcal{I}_I}$ and $\mathcal{F}_I = \sigma(\mathcal{D}_I)$. All fitted predictive quantities below are conditional on $\mathcal{F}_I$. The target is aggregate future incurred development,

$$R = \sum_{(i,j) \in \mathcal{L}_I} X_{ij}.$$

For $i = 1, \dots, I$ and $j = 0, \dots, J$, define cumulative incurred development by

$$C_{ij} = \sum_{k=0}^j X_{ik}, \quad C_{i,-1} = 0.$$

The selected volume-weighted incurred CL factor is

$$\hat{f}_j = \frac{\sum_{i=1}^{I-j-1} C_{i,j+1}}{\sum_{i=1}^{I-j-1} C_{ij}}, \quad j = 0, \dots, J-1,$$

provided the denominator is positive and finite. Note that negative incrementals are allowed. The standard Mack calculation requires positive cumulative denominators and positive selected factors. For $i = 1, \dots, I$ and $j = 0, \dots, J$, define the completed cumulative values by

$$\hat{C}_{ij}^{\text{CL}} = \begin{cases} C_{ij}, & (i, j) \in \mathcal{I}_I, \\ \hat{f}_{j-1} \hat{C}_{i,j-1}^{\text{CL}}, & (i, j) \in \mathcal{L}_I. \end{cases}$$

For $i = 1, \dots, I$ and $j = 0, \dots, J$, its incremental means are

$$\hat{\mu}_{ij}^{\text{CL}} := \hat{C}_{ij}^{\text{CL}} - \hat{C}_{i,j-1}^{\text{CL}}, \quad \hat{C}_{i,-1}^{\text{CL}} = 0.$$

The selected CL reserve sums these projected future increments. By telescoping within each accident period, it is equivalently the projected ultimate less the latest observed cumulative:

$$\hat{R}^{\text{CL}} := \sum_{(i,j) \in \mathcal{L}_I} \hat{\mu}_{ij}^{\text{CL}} = \sum_{i=1}^I \left(\hat{C}_{iJ}^{\text{CL}} - C_{i,j_i^*} \right).$$

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2.2 Mack benchmark and calibration target

The Mack CL model is used here as the deterministic incurred-CL benchmark and as an aggregate calibration target. We state its three model assumptions separately, following [Mack \(1993\)](#):

Assumption 1 (Mack CL assumptions). For $i = 1, \dots, I$ and $j = 0, \dots, J - 1$:

(MCL1) The accident-period cumulative paths $(C_{i0}, \dots, C_{iJ})$ are mutually independent across i .

(MCL2) There exist development factors $f_j > 0$ such that

$$\mathbb{E}[C_{i,j+1} \mid C_{i0}, \dots, C_{ij}] = f_j C_{ij}.$$

(MCL3) There exist variance parameters $\sigma_j^2 > 0$ such that

$$\text{Var}(C_{i,j+1} \mid C_{i0}, \dots, C_{ij}) = \sigma_j^2 C_{ij}.$$

MCL1–MCL3 define the benchmark at accident-period level. MCL1 separates accident periods, while MCL2–MCL3 specify their conditional first two moments. The selected CL completion supplies the ODS mean anchor, and the Mack MSEP supplies the aggregate calibration target.

Mack regularity conditions. For the CL and Mack formulas below, cumulative variables are non-negative, $I \geq J + 1$, and every selected development factor has at least one age-to-age pair. Whenever $C_{i,j+1}/C_{ij}$ is used, require $C_{ij} > 0$ for $i = 1, \dots, I - j - 1$ and $j = 0, \dots, J - 1$. Smaller triangles use the corresponding edge-case definitions. All other denominators and variance estimates are required to be positive and finite. For $j = 0, \dots, J - 2$, use

$$\hat{\sigma}_j^2 = \frac{1}{I - j - 2} \sum_{i=1}^{I-j-1} C_{ij} \left(\frac{C_{i,j+1}}{C_{ij}} - \hat{f}_j \right)^2.$$

In the square design used in the numerical study, $I = J + 1$. The terminal age $j = J - 1$ then has exactly one observed link ratio. In this one-ratio case, use the standard Mack extrapolation

$$\hat{\sigma}_{J-1}^2 = \min \left\{ \hat{\sigma}_{J-2}^2, \hat{\sigma}_{J-3}^2, \frac{(\hat{\sigma}_{J-2}^2)^2}{\hat{\sigma}_{J-3}^2} \right\},$$

when $I = J + 1$, $J \geq 3$, and the preceding estimates are positive ([Mack, 1993, 1999](#)). For a rectangular array with $I \geq J + 2$, the terminal age has at least two observed link ratios, and its variance coefficient is instead estimated directly by

$$\hat{\sigma}_{J-1}^2 = \frac{1}{I - J - 1} \sum_{i=1}^{I-J} C_{i,J-1} \left(\frac{C_{iJ}}{C_{i,J-1}} - \hat{f}_{J-1} \right)^2.$$

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This distinguishes the one-ratio square case from the rectangular case. The aggregate Mack MSEP used for calibration is

$$\widehat{\text{MSEP}}_{\text{Mack}}(R) = \underbrace{\sum_{i=1}^I (\hat{C}_{ij}^{\text{CL}})^2 \sum_{j=j_i^*}^{J-1} \frac{\hat{\sigma}_j^2}{(\hat{f}_j)^2 \hat{C}_{ij}^{\text{CL}}}}_{\text{process uncertainty}} + \underbrace{\sum_{j=0}^{J-1} \frac{\hat{\sigma}_j^2}{(\hat{f}_j)^2 \sum_{\ell=1}^{I-j-1} C_{\ell j}} \left(\sum_{i=I-j}^I \hat{C}_{ij}^{\text{CL}} \right)^2}_{\text{estimation uncertainty}}. \quad (1)$$

The target is used here as an aggregate prediction-error benchmark, while the ODS distribution treats the fitted parameters as plug-in values.

2.3 The ODS model

The ODS layer retains the accident-period/development-period decomposition and models each incremental cell using a sign and a nonnegative magnitude. Let

$$\theta := (\alpha_1, \dots, \alpha_I, \beta_0^+, \dots, \beta_J^+, \beta_0^-, \dots, \beta_J^-).$$

Define the cell mean by

$$\mu_{ij} := \alpha_i (\beta_j^+ - \beta_j^-).$$

The ODS distribution has integer support, so all monetary amounts are recorded as integer multiples of the chosen monetary unit. We write $\text{NB}(r, p)$ for the negative-binomial distribution with real-valued size $r > 0$ and success probability $p \in (0, 1)$. Its probability mass function is,

$$\mathbb{P}(N = n) = \frac{\Gamma(n+r)}{\Gamma(r)n!} p^r (1-p)^n, \quad n = 0, 1, 2, \dots,$$

and it has mean and variance given by

$$\mathbb{E}[N] = \frac{r(1-p)}{p}, \quad \text{Var}(N) = \frac{r(1-p)}{p^2}.$$

The ODS population and sampling assumptions are as follows:

Assumption 2 (ODS assumptions).

(ODS1) The triangle dimensions are finite, cells are integer-valued, $\varphi > 1$, $\alpha_i > 0$ for $i = 1, \dots, I$, and $\beta_j^\pm \geq 0$ for $j = 0, \dots, J$.

(ODS2) The cells are mutually independent. For each cell with $|\mu_{ij}| > 0$, its magnitude satisfies

$$N_{ij} \sim \text{NB}\left(\frac{|\mu_{ij}|}{\varphi - 1}, \frac{1}{\varphi}\right).$$

If $\mu_{ij} = 0$, set $N_{ij} = 0$ almost surely.

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Model 1 (The ODS reserving model). For $i = 1, \dots, I$ and $j = 0, \dots, J$, define the incremental cells by

$$X_{ij} := \text{sign}(\mu_{ij})N_{ij}, \quad \text{sign}(0) := 0,$$

where N_{ij} is a nonnegative magnitude as specified in ODS2 of assumption 2.

A positive-mean cell has a negative-binomial distribution, a negative-mean cell its reflection, and a zero-mean cell is a point mass at zero. The cell mean and variance are μ_{ij} and $\varphi|\mu_{ij}|$. Because $\mu_{ij} = \alpha_i(\beta_j^+ - \beta_j^-)$, for any $c > 0$ the transformation $\alpha_i \mapsto c\alpha_i$, $\beta_j^\pm \mapsto \beta_j^\pm/c$ leaves every cell mean, and hence every cell distribution, unchanged. The factors are therefore not separately identified. We impose the fitted normalisation below as a scale convention; it leaves the fitted cell means unchanged and sets $\hat{\alpha}_{iI}$ equal to the fitted CL ultimate $\hat{U}_i$:

$$\sum_{j=0}^J (\hat{\beta}_{jI}^+ - \hat{\beta}_{jI}^-) = 1.$$

Remark 1 (Mack and ODS dependence assumptions). *MCL1 allows dependence among incremental cells within an accident period, whereas ODS2 implies independence of the cells conditional on the fixed parameters. Calendar-year effects act along triangle diagonals and can induce dependence across accident periods, adding a natural extension of this development-level specification. The present ODS distribution focuses on cellwise variation while retaining the ability to represent negative incrementals in incurred data.*

2.4 Fitted ODS construction and dispersion calibration

For the fitted predictive layer, all quantities defined below are deterministic functions of $\mathcal{D}_I$ and hence $\mathcal{F}_I$ -measurable. We use φ for the population dispersion parameter and write the two data-driven calibrations as $\hat{\varphi}_{\text{Mack}}$ and $\hat{\varphi}_{\text{Pearson}}$. Thus, $\hat{\varphi}_I$ is the selected value of one of these calibrations,

$$\hat{\varphi}_I \in \{\hat{\varphi}_{\text{Mack}}, \hat{\varphi}_{\text{Pearson}}\}, \quad \hat{\varphi}_I > 1.$$

Let $\hat{\mathbb{P}}_I^{\text{ODS}}$ denote the fitted plug-in distribution for the future vector $(X_{ij})_{(i,j) \in \mathcal{L}_I}$ obtained by replacing (θ, φ) in model 1 by the fitted parameters defined below and $\hat{\varphi}_I$. The observed cells are held fixed at $\mathcal{D}_I$. We write $\widehat{\mathbb{E}}_I$ and $\widehat{\text{Var}}_I$ for moments under this distribution. It is a plug-in approximation to the conditional distribution under the original model and uses the signed negative-binomial construction in model 1. Define the CL ultimates by $\hat{U}_i := \hat{C}_{ij}^{\text{CL}}$ for $i = 1, \dots, I$. The CL-to-ODS mapping is well-defined when

$$0 < \hat{U}_i < \infty, \quad 0 < |\hat{f}_j| < \infty, \quad i = 1, \dots, I, \quad j = 0, \dots, J-1.$$

Define the development shares by

$$\hat{\gamma}_0 := \frac{1}{\prod_{k=0}^{J-1} \hat{f}_k}, \quad \hat{\gamma}_j := \frac{\hat{f}_{j-1} - 1}{\prod_{k=j-1}^{J-1} \hat{f}_k}, \quad j = 1, \dots, J.$$

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For the Mack MSEP calibration, the Mack requirements impose $\hat{f}_j > 0$ and positive cumulative denominators. In addition, at least one future cell must have positive component intensity. Otherwise the Mack MSEP denominator is zero. The absolute-value condition belongs to the ODS CL-to-ODS mapping and is imposed in addition to the Mack requirements.

Here and below, an empty product equals one and an empty sum equals zero. For every future cell $(i, j) \in \mathcal{L}_I$, so that $j \geq j_i^* + 1$, the completed CL recursion gives

$$\hat{C}_{ij}^{\text{CL}} = \frac{\hat{U}_i}{\prod_{k=j}^{j-1} \hat{f}_k}.$$

Consequently, $\hat{U}_i \hat{\gamma}_j$ is the completed CL increment at age j (whenever that age is used in the future triangle). The fitted ODS parameters are defined by

$$\hat{\alpha}_{iI} := \hat{U}_i, \quad \hat{\beta}_{jI}^+ := \max(\hat{\gamma}_j, 0), \quad \hat{\beta}_{jI}^- := \max(-\hat{\gamma}_j, 0).$$

Here $i = 1, \dots, I$ for $\hat{U}_i$ and $\hat{\alpha}_{iI}$, and $j = 0, \dots, J$ for $\hat{\gamma}_j$ and $\hat{\beta}_{jI}^\pm$.

Assumption 3 (Fitted one-sided sign structure). *The fitted predictive layer imposes $\hat{\beta}_{jI}^+ \hat{\beta}_{jI}^- = 0$ for $j = 0, \dots, J$. Thus, one signed component is active at each estimated development age. This fitted restriction concerns the estimated development shares. The population parameters in ODS1 of assumption 2 remain unrestricted.*

Consequently,

$$\hat{\mu}_{ij}^{\text{ODS}} := \hat{\alpha}_{iI} (\hat{\beta}_{jI}^+ - \hat{\beta}_{jI}^-) = \hat{U}_i \hat{\gamma}_j.$$

Because $\sum_{j=0}^J \hat{\gamma}_j = 1$, this normalisation makes $\hat{\alpha}_{iI}$ equal to the completed incurred CL ultimate. The completed CL recursion implies that $\hat{\mu}_{ij}^{\text{ODS}} = \hat{\mu}_{ij}^{\text{CL}}$ for future cells $(i, j) \in \mathcal{L}_I$. Thus the fitted ODS layer reproduces every future CL cell mean. To make the accident-year implication explicit, define the random completed ultimate for accident period i by

$$U_i^{\text{ODS}} := C_{i,j_i^*} + \sum_{j=j_i^*+1}^J X_{ij}, \quad \hat{U}_i^{\text{CL}} := \hat{C}_{ij}^{\text{CL}}.$$

Then, conditional on the observed triangle,

$$\hat{\mathbb{E}}_I[U_i^{\text{ODS}}] = C_{i,j_i^*} + \sum_{j=j_i^*+1}^J \hat{\mu}_{ij}^{\text{CL}} = \hat{U}_i^{\text{CL}}, \quad \hat{\mathbb{E}}_I[R] = \hat{R}^{\text{CL}}.$$

The fitted ODS process variance is derived in Section 3.

For the separate Pearson-type calibration, let $\hat{\mu}_{ij}^{\text{P}}$ denote the fitted observed-cell incremental mean from the age-to-age fit. In the simulation, this is the Mack CL fitted next-age cumulative value

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minus the observed current cumulative value. The future ODS distribution is determined independently of these observed-cell calibration means. The Mack MSEP in Equation (1) supplies an aggregate target. Its parameter-estimation component stays within the Mack target, while the ODS calibration adds process variation to the fitted plug-in distribution. We calibrate the fitted ODS process-variance scale using the future component-intensity scale:

$$\hat{\phi}_{\text{Mack}} = \frac{\widehat{\text{MSEP}}_{\text{Mack}}(R)}{\sum_{(i,j) \in \mathcal{L}_I} \hat{\alpha}_{iI} (\hat{\beta}_{jI}^+ + \hat{\beta}_{jI}^-)}, \quad (2)$$

provided the denominator is positive and the resulting calibration satisfies $\hat{\phi}_{\text{Mack}} > 1$. Under the fitted one-sided sign restriction in assumption 3, the denominator equals $\sum_{(i,j) \in \mathcal{L}_I} |\hat{\mu}_{ij}^{\text{CL}}|$. The coefficient is solely an aggregate calibration for the fitted plug-in distribution. Under the fitted ODS distribution and the one-sided development-period restriction, this choice gives

$$\widehat{\text{Var}}_{I,\text{proc}}(R) = \hat{\phi}_{\text{Mack}} \sum_{(i,j) \in \mathcal{L}_I} |\hat{\mu}_{ij}^{\text{CL}}| = \widehat{\text{MSEP}}_{\text{Mack}}(R),$$

whenever the calibration is admissible. This equality concerns one aggregate second moment. The Mack and ODS predictive distributions can still differ, as can their quantiles. Define the informative observed-cell set

$$\mathcal{I}_I^+ := \left\{ (i, j) \in \mathcal{I}_I : 0 < |\hat{\mu}_{ij}^{\text{P}}| < \infty, |X_{ij}| < \infty \right\}.$$

The Pearson-type dispersion estimator used in the numerical study is given below. Here, $|\mathcal{I}_I|$ denotes the number of observed incremental cells.

$$\hat{\phi}_{\text{Pearson}} = \frac{1}{\nu_{\text{P}}} \sum_{(i,j) \in \mathcal{I}_I^+} \frac{(|X_{ij}| - |\hat{\mu}_{ij}^{\text{P}}|)^2}{|\hat{\mu}_{ij}^{\text{P}}|}, \quad \nu_{\text{P}} = |\mathcal{I}_I| - (I + J). \quad (3)$$

Under the square-triangle assumption $J = I - 1$, this reduces to $\nu_{\text{P}} = |\mathcal{I}_I| - (2I - 1)$. The expression is a nominal degrees-of-freedom correction: it treats the first development column as conditioned upon and subtracts the J selected development factors. This estimator is reported only when $\nu_{\text{P}} > 0$ and $\mathcal{I}_I^+$ is nonempty and the resulting calibration satisfies $\hat{\phi}_{\text{Pearson}} > 1$. The numerator uses absolute-value residuals over informative observed cells. Unavailable fitted means are omitted, including the first development column, which is conditioned upon. A zero fitted mean is admissible only when the corresponding observed increment is also zero. A nonzero observation at a zero fitted mean makes the Pearson estimate undefined.

3 Structural properties

The following lemma derives the cell moments from the signed negative-binomial construction for $i = 1, \dots, I$ and $j = 0, \dots, J$, using the conditional cell mean μ_{ij} defined in Section 2.

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Lemma 1 (Cell moments). *Under model 1 and in particular under ODS2 in assumption 2, the cells are mutually independent and*

$$\mathbb{E}[X_{ij} \mid \theta, \varphi] = \alpha_i(\beta_j^+ - \beta_j^-) =: \mu_{ij}, \quad \text{Var}(X_{ij} \mid \theta, \varphi) = \varphi|\mu_{ij}|.$$

Proof. For $|\mu_{ij}| > 0$, N_{ij} has shape $|\mu_{ij}|/(\varphi - 1)$ and probability $1/\varphi$. Its mean is $|\mu_{ij}|$ and its variance is $\varphi|\mu_{ij}|$. Multiplication by $\text{sign}(\mu_{ij})$ gives mean μ_{ij} and leaves the variance unchanged. The zero-mean point-mass convention gives the same formulas. $\square$

Recall from Section 2 that R is aggregate future incurred development.

Proposition 1 (ODS aggregate distribution and moments). *Under assumption 2, conditionally on fixed (θ, φ) with $\varphi > 1$, define the positive and negative future-mean totals*

$$M_+ := \sum_{(i,j) \in \mathcal{L}_I: \mu_{ij} > 0} \mu_{ij}, \quad M_- := \sum_{(i,j) \in \mathcal{L}_I: \mu_{ij} < 0} |\mu_{ij}|.$$

Let R_+ and R_- be the sums of future cell magnitudes on the positive-mean and negative-mean cells, respectively. Then R_+ and R_- are independent, with

$$R_+ \sim \text{NB}\left(\frac{M_+}{\varphi - 1}, \frac{1}{\varphi}\right), \quad R_- \sim \text{NB}\left(\frac{M_-}{\varphi - 1}, \frac{1}{\varphi}\right).$$

When $M_{\pm} = 0$, interpret the corresponding total as zero almost surely. Since future cells with $\mu_{ij} = 0$ are also zero almost surely, $R = R_+ - R_-$. Its mean and variance are

$$\mathbb{E}[R \mid \theta, \varphi] = M_+ - M_- = \sum_{(i,j) \in \mathcal{L}_I} \mu_{ij}, \quad \text{Var}(R \mid \theta, \varphi) = \varphi(M_+ + M_-) = \varphi \sum_{(i,j) \in \mathcal{L}_I} |\mu_{ij}|.$$

Proof. By ODS2, nonzero-mean cell magnitudes are independent negative-binomial variables with common $p = 1/\varphi$. Zero-mean cells are zero almost surely. The positive- and negative-mean sets are disjoint, so their sums are independent. Convolution adds their shape parameters, giving the stated distributions. Their negative-binomial moments give the displayed mean and variance. $\square$

In the quasi-likelihood sense, the aggregate has the ODS mean-variance structure: compared with a classical Skellam variable with component means M_+ and M_- , its mean is unchanged and its variance is multiplied by φ (Skellam, 1946). The fitted distribution has the same aggregate representation after replacing unknown parameters by fitted values. In particular, by the construction in Section 2,

$$\widehat{\mathbb{E}}_I[X_{ij}] = \hat{\mu}_{ij}^{\text{CL}}, \quad (i, j) \in \mathcal{L}_I,$$

and hence

$$\widehat{\mathbb{E}}_I[R] = \hat{R}^{\text{CL}}, \quad \widehat{\text{Var}}_{I,\text{proc}}(R) = \hat{\varphi}_I \sum_{(i,j) \in \mathcal{L}_I} |\hat{\mu}_{ij}^{\text{CL}}|.$$

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For the accident-year ultimate defined in Section 2,

$$\hat{\mathbb{E}}_I[U_i^{\text{ODS}}] = \hat{U}_i^{\text{CL}}, \quad i = 1, \dots, I.$$

The fitted ODS distribution preserves the selected CL mean surface and gives the aggregate process variance stated above. The aggregate representation supports two simulation methods. One method simulates future cells independently. The other simulates the two aggregate negative-binomial components directly.

Proposition 2 (ODP moment special case). *If $\beta_j^- = 0$ for $j = 0, \dots, J$, then for $i = 1, \dots, I$ and $j = 0, \dots, J$,*

$$\mathbb{E}[X_{ij} \mid \theta, \varphi] = \alpha_i \beta_j^+, \quad \text{Var}(X_{ij} \mid \theta, \varphi) = \varphi \alpha_i \beta_j^+,$$

which is the classical one-sided over-dispersed Poisson (ODP) mean-variance structure. The ODP model is therefore the one-sided special case of the ODS model ([England and Verrall, 2002](#)).

4 Numerical study

Appendix A specifies the complete data-generating process (DGP), which follows the ODS assumptions. The study evaluates prediction intervals for the aggregate future incurred development $R = \sum_{(i,j) \in \mathcal{L}_I} X_{ij}$, using the observed upper triangle to fit the incurred CL mean surface. Conditional on the fixed DGP parameters, the incremental cells are independent. Note that the DGP is not a full Mack-generating model ([Mack, 1993](#)). Mack supplies conditional moment assumptions for the CL benchmark and MSEP calibration without imposing independence among increments within an accident period.

All three predictive distributions use the fitted incurred CL means and differ only in the dispersion used for future cells: the oracle process-only benchmark uses the known DGP dispersion, the Mack-MSEP method uses the aggregate Mack MSEP calibration, and the Pearson method uses the observed-cell Pearson calibration. Figure 1 provides a visual preview of the simulation design: the expected cumulative triangle and the corresponding CL development factors. The early negative development pattern reflects incurred reserve releases, followed by positive later development. The parameter values and cell-generation mechanism are given in Appendix A.

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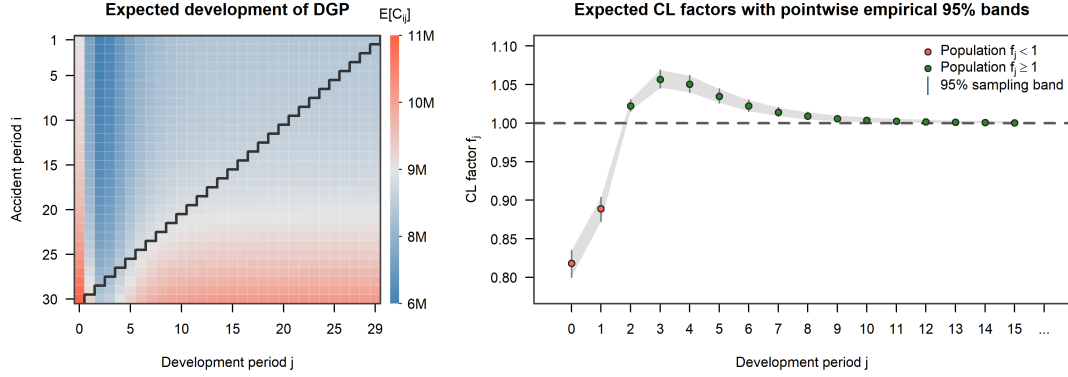


Figure 1: Overview of the simulation DGP: expected cumulative development and CL factors.

4.1 Simulation protocol

For each of $S = 10\,000$ outer replications, we simulate a full incremental triangle under Appendix A. We retain the observed cells in $\mathcal{I}_I$, regard the cells in $\mathcal{L}_I$ as future, and record the realised future reserve $R^{(s)}$. We then fit incurred Mack CL to the observed triangle and obtain its fitted future mean surface. The oracle process-only benchmark uses this fitted centre together with the known DGP dispersion φ_{DGP} . It is infeasible because that dispersion is not estimated from the observed triangle. The two feasible alternatives use the same fitted centre with, respectively, the Mack-MSEP calibration $\hat{\varphi}_{\text{Mack}}$ and the observed-cell Pearson calibration $\hat{\varphi}_{\text{Pearson}}$.

For each valid fitted replication and each of the three dispersion choices, we simulate $B = 10\,000$ conditionally independent future-cell realisations under the ODS distribution and sum the future cells to obtain reserve draws. A replication is valid only when both feasible dispersion estimates are finite and admissible. Equal-tailed empirical intervals use probabilities (2.5%, 97.5%) and (0.5%, 99.5%) for nominal 95% and 99% coverage, respectively. Coverage is recorded when the interval contains the realised reserve $R^{(s)}$.

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Algorithm 1 Monte Carlo protocol for coverage assessment

Input: Simulation design in Appendix A. Set $S = 10\,000$ and $B = 10\,000$.

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1: for  $s = 1, \dots, S$  do
2:   Simulate a full triangle, retain  $\mathcal{I}_I$ , and record  $R^{(s)}$  over  $\mathcal{L}_I$ 
3:   Fit incurred Mack CL and obtain the fitted future mean surface
4:   Calculate  $\hat{\phi}_{\text{Mack}}^{(s)}$  and  $\hat{\phi}_{\text{Pearson}}^{(s)}$ 
5:   if both feasible estimates are valid then
6:     for each  $\phi \in \{\varphi_{\text{DGP}}, \hat{\phi}_{\text{Mack}}^{(s)}, \hat{\phi}_{\text{Pearson}}^{(s)}\}$  do
7:       Simulate  $B$  future reserves from the fitted ODS mean surface and dispersion  $\phi$ 
8:       Construct 95% and 99% intervals and record coverage indicators
9:     end for
10:  end if
11: end for
12: Report coverage and interval-width summaries over valid replications

```

4.2 Results

The point-estimation summaries appear in Table 1. The DGP oracle row reports the fixed DGP expected reserve and known dispersion and is thus not a fitted method. The two feasible methods use the same incurred CL centre, so their average fitted reserves and average differences from the fixed DGP expected reserve are equal. The Mack-MSEP calibration uses the component-intensity scale over $\mathcal{L}_I$ to match the aggregate Mack MSEP. The Pearson calibration uses observed-cell residuals over $\mathcal{I}_I$ and targets a separate dispersion quantity. Neither method adds a separate parameter-draw layer: the fitted mean surface and calibrated dispersion are plug-in values in each outer replication.

Method	Average reserve	Difference from DGP	Dispersion used
DGP oracle	7 322 061		100 000
Mack-MSEP	7 320 764	−1 297 (−0.02%)	172 417
Pearson-based	7 320 764	−1 297 (−0.02%)	7 851 377

Table 1: Method-wise average reserve, difference from the DGP expected reserve, and dispersion used. Note that the DGP oracle uses the fixed DGP expected reserve and known dispersion.

All predictive methods use the same fitted incurred CL mean surface. The common average fitted reserve is 7 320 764, close to the fixed DGP expected reserve of 7 322 061, with a difference of −0.02%. Thus, the mean centre is adequately captured in this simulation. However, the dispersion under Pearson calibration is materially larger than under the other calibrations.

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Prediction-interval coverage. Figure 2 illustrates the empirical coverage and interval widths. All intervals are conditional on the fitted CL centre within an outer replication. Process-only intervals use the known DGP dispersion, so their inner draws represent future process variation but do not propagate uncertainty from estimating the dispersion or the fitted mean. Across outer replications, coverage nevertheless reflects both the displacement of the fitted centre and future process variation. In this experiment, process-only intervals under-cover, Mack-MSEP intervals are close to nominal, and Pearson intervals are much wider and over-cover. These are design-specific empirical coverage results, not unconditional claims about all incurred-reserving portfolios.

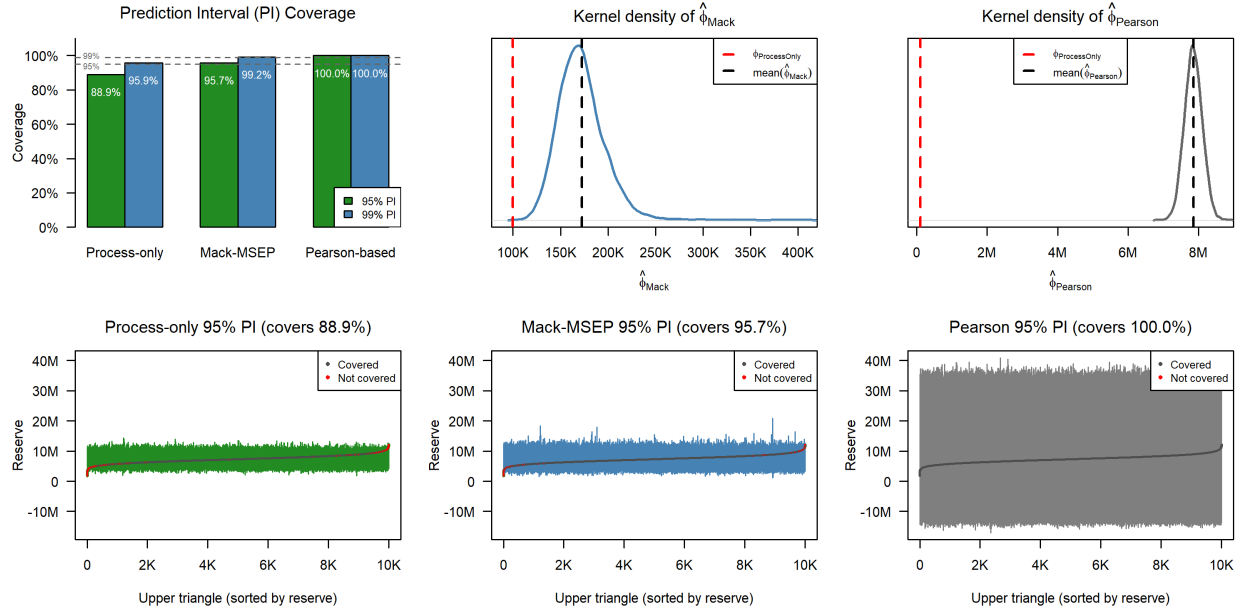


Figure 2: Distributions of the two feasible dispersion calibrations and 95% prediction intervals under process-only, Mack-MSEP, and Pearson-based dispersion.

5 Discussion

The practical role of ODS is to represent predictive uncertainty conditional on a selected incurred completion, not to select or validate the economically and legally appropriate best-estimate reserve. This distinction is relevant under Solvency II: Article 272(1)(b) of Commission Delegated Regulation (EU) 2015/35 ([European Commission, 2015](#)) requires the actuarial function to assess the uncertainty associated with the estimates made in the calculation of technical provisions. ODS may support this assessment where signed incurred development is material. The best-estimate reserve and its decomposition into RBNS, IBNR, and future payments must be established separately. The predictive layer also requires a specified development pattern for future incurred movement around the selected completion. Its suitability depends on portfolio-specific validation of dependence, calibration, and model assumptions.

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The ODS layer could also be formulated around a selected BF completion. A selected BF ultimate and development pattern imply an incremental mean surface that could replace the selected CL incremental mean surface in Section 2, subject to the stated sign and admissibility conditions. An analogous dispersion calibration could use a BF MSEP estimator (Saluz, 2011). For comparison, a recent preprint derives an analytical MSEP for a stochastic lift of the GCC method (Richman and Wüthrich, 2026). GCC could therefore provide an alternative mean–MSEP framework for future comparison, whereas ODS provides the signed predictive distribution considered here. Numerical evaluation of the BF extension and its comparison with GCC are left to future work. The conditional formulation could apply to annual, quarterly, or monthly development periods. It can serve as a stochastic layer in direct best-estimate projection, latent IBNR modelling, and broader incurred-reserving applications.

Scope and limitations. The fitted one-sided sign restriction gives one active component at each future development age, although signs may differ across ages. The general cell model permits both components to be active simultaneously. ODS2 in assumption 2 specifies mutual independence of all cell components, and hence cell independence conditional on the fixed model parameters. Calendar-year effects and common shocks are currently outside the present model, but other ODP/CL extensions address that separate issue (Brydon and Verrall, 2009; Wüthrich, 2010; Riegel, 2026). Under the fitted one-sided restriction, the scalar dispersion multiplier gives

$$\widehat{\text{Var}}_{I,\text{proc}}(X_{ij}) = \hat{\phi}_I |\hat{\mu}_{ij}^{\text{CL}}|, \quad (i, j) \in \mathcal{L}_I.$$

A development-specific extension can split an additional component intensity $\kappa_j \geq 0$ equally between the two fitted component shares:

$$\tilde{\beta}_j^+ = \hat{\beta}_{jI}^+ + \frac{\kappa_j}{2}, \quad \tilde{\beta}_j^- = \hat{\beta}_{jI}^- + \frac{\kappa_j}{2}.$$

With $\tilde{\lambda}_{ij}^\pm = \hat{\alpha}_{iI} \tilde{\beta}_j^\pm$, the fitted future means remain unchanged,

$$\tilde{\mu}_{ij} = \hat{\alpha}_{iI} (\tilde{\beta}_j^+ - \tilde{\beta}_j^-) = \hat{\mu}_{ij}^{\text{ODS}}, \quad (i, j) \in \mathcal{L}_I,$$

whereas the predictive variance becomes

$$\text{Var}_\kappa(X_{ij} \mid \mathcal{F}_I) = \hat{\phi}_I \hat{\alpha}_{iI} \left(\hat{\beta}_{jI}^+ + \hat{\beta}_{jI}^- + \kappa_j \right), \quad (i, j) \in \mathcal{L}_I.$$

Thus κ_j preserves the fitted means while increasing predictive variance. Its estimation and calibration as well as calendar-year dependence and parameter uncertainty require further validation.

6 Conclusion

We develop ODS as a conditional predictive layer for future incurred development around a selected incurred chain-ladder completion. Whenever the fitting conditions are met, it preserves

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the selected CL mean in every future cell and therefore preserves the corresponding expected aggregate reserve and accident-year ultimates. The deterministic reserving centre is consequently separated from the stochastic distribution used to represent future variation.

The Mack MSEP calibration matches the aggregate second moment of the fitted ODS process variance. Distributional features, quantiles, and parameter- uncertainty treatment remain model-specific. The Pearson-type calibration targets a different quantity and is interpreted separately.

In the stated simulation, all methods captured the mean adequately because they shared the same fitted CL centre, while their uncertainty assessments differed: the process-only benchmark under-covered, the Mack MSEP calibration was close to nominal, and the Pearson calibration produced wider intervals and over-covered. These findings are specific to the stated data-generating process and motivate validation under alternative portfolios and calibration choices.

ODS is therefore a transparent actuarial modelling option for conditional future incurred development, subject to portfolio-specific validation of its mean structure, sign restriction, dispersion calibration, and dependence assumptions. Its results provide model-based predictive assessments to be used alongside validation and actuarial judgement.

A Simulation study details

We introduce the DGP used for the numerical experiment. We set $I = J + 1 = 30$ and generate a full 30×30 incremental triangle with independent cells, as in ODS3–ODS4 of assumption 2. The cell mean is

$$\mu_{ij} = \alpha_i(\beta_j^+ - \beta_j^-), \quad i = 1, \dots, 30, \quad j = 0, \dots, 29. \quad (4)$$

The ultimate amounts α_i increase from 8 million to 10 million in the chosen monetary unit. Let $\gamma_j = \beta_j^+ - \beta_j^-$. We set

$$\gamma_0 = 1.1, \quad \gamma_1 = -0.2, \quad \gamma_2 = -0.1,$$

and allocate the remaining mass 0.2 over later development periods according to a discretised log-normal shape, so that $\sum_j \gamma_j = 1$. Define

$$\beta_j^+ = \max(\gamma_j, 0), \quad \beta_j^- = \max(-\gamma_j, 0).$$

Thus, development periods 1 and 2 have negative net mean increments.

For $\varphi_{\text{DGP}} = 100\,000$, each cell is generated as

$$X_{ij} = Y_{ij}^+ - Y_{ij}^-, \quad Y_{ij}^\pm \sim \text{NB} \left(\frac{\alpha_i \beta_j^\pm}{\varphi_{\text{DGP}} - 1}, \frac{1}{\varphi_{\text{DGP}}} \right), \quad (5)$$

with independent components and the point-mass convention for a zero component. The value of φ_{DGP} is tied to the chosen integer monetary unit. The DGP is therefore an instance of the ODS model, with negative early incurred development and positive later development. A visual overview is provided in Figure 1 in the numerical study.

Generative AI declaration

Generative AI was used for coding assistance and formatting. The author takes full responsibility for the final content.

References

- Brydon, D. and R. J. Verrall (2009). 'Calendar year effects, claims inflation and the chain-ladder technique'. *Annals of Actuarial Science* **4**(2), 287–301.
- de Alba, E. (2006). 'Claims reserving when there are negative values in the runoff triangle: Bayesian analysis using the three-parameter log-normal distribution'. *North American Actuarial Journal* **10**(3), 45–59.
- England, P. and R.J. Verrall (2002). 'Stochastic claims reserving in general insurance'. *British Actuarial Journal* **8**(3), 443–518.
- England, P. and R.J. Verrall (2006). 'Predictive distributions of outstanding liabilities in general insurance'. *Annals of Actuarial Science* **1**(2), 221–270.
- European Commission (2015). 'Commission delegated regulation (eu) 2015/35 of 10 october 2014 supplementing directive 2009/138/ec of the european parliament and of the council on the taking-up and pursuit of the business of insurance and reinsurance (solvency ii)'. Official Journal of the European Union, L 12, 17 January 2015, pp. 1–797. EUR-Lex: https://eur-lex.europa.eu/eli/reg_del/2015/35/oj/eng.
- Mack, T. (1993). 'Distribution-free calculation of the standard error of chain-ladder reserve estimates'. *ASTIN Bulletin* **23**(2), 213–225.
- Mack, T. (1994). 'Which stochastic model is underlying the chain ladder method?'. *Insurance: Mathematics and Economics* **15**(2–3), 133–138.
- Mack, T. (1999). 'The standard error of chain ladder reserve estimates: Recursive calculation and inclusion of a tail factor'. *ASTIN Bulletin* **29**(2), 361–366.
- Renshaw, A. E. and Richard J. Verrall (1998). 'A stochastic model underlying the chain-ladder technique'. *British Actuarial Journal* **4**(4), 903–923.
- Richman, R. and Mario V. Wüthrich (2026). 'A note on the generalized cape cod reserving method'.
- Riegel, U. (2026). 'A chain ladder method using prior information on calendar year effects'. *ASTIN Bulletin* pp. 1–28.
- Saluz, A. (2011). 'Development pattern and prediction error for the stochastic bornhuetter-ferguson claims reserving method'. *ASTIN Bulletin* **41**(2), 279–313.
- Scarth, R., Saanya Jain and Rocco Roberto Cerchiara (2020). *A Practitioner's Introduction to Stochastic Reserving - The One-Year View*. Institute and Faculty of Actuaries.

The Over-Dispersed Skellam (ODS) Reserving Model

- Skellam, J. G. (1946). 'The frequency distribution of the difference between two poisson variates belonging to different populations'. *Journal of the Royal Statistical Society. Series A (General)* **109**(3), 296.
- Verrall, R. J. and Zhi Li (1993). 'Negative incremental claims: chain ladder and linear models'. *Journal of the Institute of Actuaries* **120**(1), 171–183.
- Wüthrich, M. V. (2010). 'Accounting year effects modeling in the stochastic chain ladder reserving method'. *North American Actuarial Journal* **14**(2), 235–255.
- Wüthrich, M. V. and Michael Merz (2008). *Stochastic claims reserving methods in insurance*. Wiley finance series. John Wiley & Sons.